

Nate Faulkenberry

Glastonbury, CT · nate.faulkenberry@gmail.com · (860) 710-0897
github.com/NateFaulkenberry · natefaulkenberry.github.io/portfolio

Senior Software Engineer · Full-Stack Web Development · Application Architecture · AI-Assisted Software Engineering

SUMMARY

Senior software engineer with extensive experience designing, building, modernizing, and maintaining full-stack web applications across higher education, enterprise software, consulting, and independent development.

Experienced across the full software development lifecycle, including requirements analysis, technical design, application architecture, implementation, testing, deployment, production support, and legacy modernization. Strong background in TypeScript, JavaScript, React, Next.js, Node.js, REST APIs, SQL, Docker, AWS, and modern web application architecture.

In recent years, developed a structured approach to AI-assisted software engineering, incorporating coding agents into implementation, codebase analysis, testing, debugging, refactoring, documentation, and larger multi-agent development workflows.

CORE TECHNICAL SKILLS

Languages & Frameworks

TypeScript · JavaScript · React · Next.js · Node.js · HTML5 · CSS3 · PHP · Java · SQL

Backend & Data

REST / JSON APIs · PostgreSQL · MySQL · Prisma · tRPC · Redis · AWS S3 · ETL / Data Integration · Third-Party API Integration

Architecture & Engineering

Full-Stack Application Architecture · Requirements Analysis · Technical Design · Legacy Modernization · Refactoring · Performance Optimization · Automated Testing · Production Support · Technical Documentation

Infrastructure & Development

Docker · Git · GitHub · GitHub Actions · CI/CD · AWS · Heroku · Boomi · Visual Studio Code

AI-Assisted Software Engineering

AI Coding Agents · Agentic Development Workflows · Specification-Driven Development · AI-Assisted Implementation · Legacy Code Transformation · AI-Assisted Testing · Codebase Analysis · Debugging · Multi-Agent Orchestration · Repository Automation

Additional Technologies

Bootstrap · WordPress · jQuery · Selenium · QUnit · Adobe Creative Suite

PROFESSIONAL EXPERIENCE

YALE UNIVERSITY – SCHOOL OF MANAGEMENT

Software Engineer 3 · New Haven, CT

May 2019 – Present

Full-stack software engineer responsible for designing, building, deploying, and maintaining production applications supporting faculty, staff, students, and academic operations.

Progressed from primarily frontend development into full-stack engineering and application ownership, working across architecture, backend services, databases, APIs, frontend development, testing, deployment, and production support.

Work ranges from independent development to collaboration with small cross-functional engineering teams. Responsibilities include translating ambiguous requirements and existing application behavior into technical designs, selecting appropriate technologies and architecture, implementing new systems, modernizing legacy applications, and supporting production systems throughout their lifecycle.

Course Auction

Full modernization of a legacy course auction platform

Led the modernization of a long-running course auction application originally built with .NET and Microsoft SQL Server. The system supports student course bidding through multiple auction stages and includes substantial administrative and business-rule complexity.

- Reverse-engineered a legacy MVC application with limited formal documentation, working with existing code and stakeholders to reconstruct application behavior and business requirements.

- Rebuilt the application using **Next.js, React, TypeScript, PostgreSQL, Prisma, and tRPC**, creating a strongly typed application stack from database through frontend.
- Recreated and redesigned major workflows covering auction configuration, course and student imports, permissions, bidding rounds, course availability, waitlists, and student bid management.
- Led implementation and coordinated structured testing through a realistic multi-month mock auction designed to exercise normal workflows and edge cases.
- Oversaw the first production launch of the rewritten application during a live academic auction.
- Introduced AI-assisted development into the modernization effort, using coding agents for legacy code transformation, React component generation, database schema development, project scaffolding, testing, and general implementation.
- Established a development methodology combining AI-assisted implementation with explicit specifications, architectural decisions, automated testing, regression protection, and human review.

Yale Directory

Modern replacement for an externally developed school directory

Designed and built a replacement for an existing directory application with performance and functionality limitations. The application provides directory information and profile assets to the School of Management community.

- Designed and implemented the application using **Next.js, React, TypeScript, Node.js, Docker, Redis, PostgreSQL, and AWS S3**.
- Designed a Redis caching layer between the application and legacy database to reduce repeated database access and improve application responsiveness.
- Designed an asynchronous update queue using Node.js to prevent slow database operations from blocking user-facing requests.
- Built custom fuzzy name search using **Jaro-Winkler distance**, providing autocomplete and tolerance for misspellings and alternate name forms.
- Designed and implemented the profile-image pipeline including validation, client-side cropping, compression, face detection, thumbnail generation, and S3 storage.
- Integrated TensorFlow-based face detection into automated profile-photo processing.
- Developed the application as a solo engineering project over approximately one year.
- Established reusable patterns around Docker, Next.js, React, Node.js, TypeScript, and the school's Bootstrap-based design system for subsequent applications.

- Developed a prototype for natural-language directory search integrating embeddings, vector search, LangChain, LLM integration, query translation, and the application interface.

Event Registration

Evolution of an internal event-registration platform into a generalized dynamic form system

- Redesigned the frontend of an internal event-registration application and expanded it from an event-specific form generator into a generalized dynamic form platform.
- Built the frontend integration with middleware responsible for synchronizing registration data with **Salesforce Education Cloud**.
- Collaborated with Salesforce and PHP middleware developers to deliver the end-to-end system.
- Later returned to the project to refactor the frontend into a modern **Next.js** application running in Docker.

Faculty Website Migration

Faculty WordPress migration and platform transition

- Took primary ownership of migrating faculty websites to a new WordPress environment.
- Completed content migration and resolved issues across migrated sites while working directly with faculty and support staff.
- Created documentation and conducted training for support staff.
- Supported the transition from the previous environment into the new platform.

Peek

Classroom digital-signage platform

- Rebuilt a classroom digital-signage application using **Next.js and React**.
- Provides current classroom schedules and event status across the school's classrooms.
- Application remains in production.

Selected Additional Yale Applications

- **Course Roster** — Internal faculty and staff application using directory data and assets to display and print course roster information including names, photos, biographies, and contact information.

- **Zoom Background Generator** — Browser-based application using html2canvas to generate branded Zoom backgrounds from approved institutional imagery, fonts, colors, and logos. Published publicly for use by other organizations.
- **TA Registration** — Developed frontend functionality for faculty to register and manage teaching assistants through Salesforce.
- **EMBA Course Registration** — Built frontend functionality for executive MBA students to select semester courses while enforcing program-specific business rules.
- **Canvas Syllabus Link Sync** — Developed a Boomi ETL workflow extracting Canvas information and creating syllabus links in the course database.
- **COVID-19 Data Application** — Led frontend development for an application displaying research data from Professor Matthew Spiegel's COVID-19 research project during 2020.

CAGE JSA

Web Consultant · 2017 – 2019

Provided web development and consulting services for clients, primarily building and maintaining WordPress websites, marketing funnels, and supporting web functionality.

- Worked directly with clients to translate business and marketing requirements into functional web experiences.
- Designed and developed WordPress websites, landing pages, and marketing funnels.
- Provided frontend development, troubleshooting, maintenance, and technical consulting.
- Worked across multiple client environments and adapted solutions to different requirements and workflows.

ENVISION PHARMA GROUP

Software Developer I · 2014 – 2017

- Developed and maintained software supporting an enterprise pharmaceutical technology platform.
- Implemented functionality across frontend and backend systems within established application architectures.
- Collaborated with developers, designers, and business stakeholders to translate requirements into production software.
- Supported applications through development, testing, deployment, and ongoing maintenance.

Front-End Web Developer · 2010 – 2014

- Developed interfaces for enterprise benefits administration applications.
- Built reusable frontend functionality using JavaScript, HTML, and CSS.
- Collaborated with backend developers, designers, and business stakeholders.
- Enhanced and maintained production applications while delivering new functionality.

ULTIMATE NUTRITION

Web Developer · 2010

Developed and maintained web functionality for an online nutrition and e-commerce business.

EPATH LEARNING

Pro Services Web / Flash Designer & Developer · 2008 – 2010

Developed custom web and interactive learning experiences for clients, working across design, development, and client delivery.

AI-ASSISTED SOFTWARE ENGINEERING

Beginning in 2025, developed a structured methodology for incorporating AI coding agents into professional and independent software development.

- Use AI coding agents as part of implementation, codebase exploration, legacy modernization, debugging, refactoring, documentation, and test development.
- Orchestrate multiple coding agents on larger development tasks by decomposing complex objectives into independent research, implementation, testing, and review work.
- Structure repositories and development processes to support effective long-running agent workflows through clear architecture, repository conventions, architectural decision records, implementation specifications, test plans, and regression suites.
- Use agents for large-codebase exploration, repository-wide refactoring, implementation, documentation, and automated testing.

- Develop supporting automation including repository tooling, GitHub Actions, development scripts, and reusable project instructions for agent workflows.
- Select AI models and workflows according to task complexity rather than relying on a single model or development pattern.
- Treat agent-assisted development as an engineering process involving objective definition, task decomposition, constraints, implementation, testing, review, and iterative correction.

SELECTED INDEPENDENT SOFTWARE PROJECTS

Independent software projects demonstrating continued technical development across native C++, audiovisual systems, audio software, automation, and AI-assisted engineering.

AV Gen

Native C++ Audiovisual Application

Building a real-time audiovisual application in **C++** spanning audio analysis, procedural environments, animation, GPU-accelerated rendering, interactive 3D systems, and generative audiovisual content.

The project incorporates a substantial automated testing infrastructure and multiple major subsystems covering the complete pipeline from live audio input through analysis, modulation, scene generation, and rendering.

PX3 Synth

Native C++ Software Synthesizer and Audio Plugin

Developing and publicly releasing a native C++ software synthesizer and audio plugin for **Apple Logic**, including synthesis, modulation, effects, automation, preset management, and production-oriented plugin workflows.

Engineering Automation

Developing automated development, testing, build, and deployment workflows using **GitHub Actions**, repository tooling, development scripts, and related automation.

AI-Assisted Engineering

Applying specification-driven development, architectural documentation, coding agents, automated testing, repository automation, and multi-agent workflows to substantial independent software projects.

Additional Development

Exploring and developing projects using **React Native**, **Flutter**, **Electron**, **browser technologies**, **automation utilities**, and **music-theory applications**.

EDUCATION

NEW ENGLAND INSTITUTE OF TECHNOLOGY

Associate Degree, Multimedia & Web Design · 2007